Case Study:

(1) Write a program to accept the marks of 10 students and find the average, maximum and minimum marks.

Structures

12.5 Structures Introduction

Structure is a user-defined which has the combination of data items with different data types. This allows to group variables of mixed data types together into a single unit.

12.5.1 Purpose of Structures

In any situation when more than one variable is required to represent objects of uniform data-types, array can be used. If the elements are of different data types,then array cannot support. If more than one variable is used, they can be stored in memory but not in adjacent locations. It increases the time consumption while searching. The structure provides a facility to store different data types as a part of the same logical element in one memory chunk adjacent to each other.

12.5.2 Declaring and defining structures

Structure is declared using the keyword 'struct'. The syntax of creating a structure is given below.

```
struct structure_name {
        type member_name1;
        type member_name2;
} reference_name;
```

Objects declared along with structure definition are called global objects

An optional field reference_name can be used to declare objects of the structure type directly.

Example:

```
struct Student
{
        long rollno;
        int age;
        float weight;
} ;
```

In the above declaration of the struct, three variables rollno,age and weight are used. These variables(data element)within the structure are called members (or fields). In order to use the Student structure, a variable of type Student is declared and the memory allocation is shown in figure 12.5

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